

Revised Generation Interconnection System Impact Study Report

For Queue Project Q15-0526

Southwood 138 kV
March 28, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0526. The Interconnection Customer proposes a 400.6 MW natural gas combined cycle facility interconnecting at the Southwood 138 kV substation in the Kentucky area. The request was received on October 20, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 249 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$174,670,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Skeena - Pin Hook 138 kV
Contingency: Barnabie Cable Sta - Caino 138 kV line
Loading: 274.4% (Rating: 129 MVA)
DFAX: 35.6%
Pre-project loading: 163.8%
2. Facility: Rapids Energy Center - Casino 138 kV L
Contingency: Cedar Tap - Terry Co 138 kV
Loading: 213.5% (Rating: 239 MVA)
DFAX: 54.4%
Pre-project loading: 122.3%
3. Facility: Campbell Junction Substation - Rapids Energy Center 138 kV
Contingency: Cedar Tap - Terry Co 138 kV
Loading: 212.4% (Rating: 230 MVA)
DFAX: 54.4%
Pre-project loading: 117.5%
4. Facility: Casno - Barnabie Cable Sta 138 kV
Contingency: Terry Co - Cedar Tap 138 kV
Loading: 210.2% (Rating: 248 MVA)
DFAX: 54.4%
Pre-project loading: 122.2%
5. Facility: Wixom - Albion Rd Substation 138 kV
Contingency: Casino - Barnabie Cable Sta 138 kV L
Loading: 201.8% (Rating: 255 MVA)
DFAX: 33.7%
Pre-project loading: 149.0%

6. Facility: Skeena - Terry Co 138 kV line
Contingency: Barnabie Cable Sta - Casino 138 kV
Loading: 193.8% (Rating: 250 MVA)
DFAX: 48.9%
Pre-project loading: 115.5%
7. Facility: Cedar Tap - Terry Co 138 kV line
Contingency: Barnabie Cable Sta - Casino 138 kV
Loading: 192.1% (Rating: 254 MVA)
DFAX: 48.9%
Pre-project loading: 115.1%
8. Facility: Anson N - Pin Hook 138 kV
Contingency: Barnabie Cable Sta - Casino Substation 138 kV
Loading: 179.8% (Rating: 191 MVA)
DFAX: 35.6%
Pre-project loading: 105.1%
9. Facility: Barnabie Cable Sta - Moss Bluff 138 kV
Contingency: Longdon - Barnabie Cable Sta 138 kV
Loading: 165.1% (Rating: 238 MVA)
DFAX: 59.2%
Pre-project loading: 65.6%
10. Facility: Albion Rd - Platte City 138 kV
Contingency: Albion Rd - Central Campus Unm 138 kV line
Loading: 151.2% (Rating: 238 MVA)
DFAX: 20.5%
Pre-project loading: 116.6%
11. Facility: C-33 Doe - Platte City 138 kV
Contingency: Central Campus Unm - Albion Rd 138 kV
Loading: 149.0% (Rating: 234 MVA)
DFAX: 20.5%
Pre-project loading: 113.8%
12. Facility: C-33 Doe - Sand Springs 138 kV
Contingency: Central Campus Unm - Albion Rd 138 kV
Loading: 148.1% (Rating: 229 MVA)
DFAX: 20.5%
Pre-project loading: 112.2%
13. Facility: Longdon - Barnabie Cable Sta Substation 138 kV line
Contingency: Moss Bluff - Barnabie Cable Sta 138 kV L
Loading: 146.4% (Rating: 217 MVA)
DFAX: 22.5%
Pre-project loading: 104.9%
14. Facility: Glendale S Tap 6 - Campbell Junction 138 kV
Contingency: Cedar Tap Substation - Terry Co 138 kV
Loading: 144.8% (Rating: 242 MVA)
DFAX: 36.7%
Pre-project loading: 84.0%
15. Facility: Biscoe - Glendale South Tap 6 138 kV
Contingency: Cedar Tap - Terry Co 138 kV
Loading: 142.2% (Rating: 231 MVA)
DFAX: 36.7%
Pre-project loading: 78.6%
16. Facility: Biscoe - Anson N 138 kV L
Contingency: Cedar Tap Substation - Terry Co 138 kV

- Loading: 139.1% (Rating: 223 MVA)
DFAX: 36.7%
Pre-project loading: 73.2%
17. Facility: Henry Creek - Central Campus Unm 138 kV
Contingency: Albion Rd - Platte City 138 kV Line
Loading: 133.5% (Rating: 177 MVA)
DFAX: 13.3%
Pre-project loading: 103.4%
18. Facility: Lovett - Idaho Falls Power 138 kV Line
Contingency: Barnabie Cable Sta - Casino 138 kV
Loading: 132.6% (Rating: 229 MVA)
DFAX: 37.1%
Pre-project loading: 67.7%
19. Facility: Wec - Wellington 138 kV
Contingency: Barnabie Cable Sta - Casino 138 kV
Loading: 132.1% (Rating: 243 MVA)
DFAX: 34.4%
Pre-project loading: 75.5%
20. Facility: Lovett - Barnabie Cable Sta 138 kV Line
Contingency: Casino - Barnabie Cable Sta Substation 138 kV
Loading: 131.5% (Rating: 193 MVA)
DFAX: 33.8%
Pre-project loading: 61.4%
21. Facility: Idaho Falls Power - Cedar Tap 138 kV
Contingency: Barnabie Cable Sta Substation - Casino 138 kV L
Loading: 130.3% (Rating: 237 MVA)
DFAX: 37.1%
Pre-project loading: 67.6%
22. Facility: Saint Joseph - Chestnut 138 kV Line
Contingency: Albion Road - Wixom 138 kV
Loading: 126.9% (Rating: 202 MVA)
DFAX: 29.4%
Pre-project loading: 68.5%
23. Facility: Albion Road - Central Campus Unm 138 kV
Contingency: Albion Rd - Platte City Substation 138 kV
Loading: 126.5% (Rating: 195 MVA)
DFAX: 13.3%
Pre-project loading: 99.2%
24. Facility: Davisille - N Appleton 138 kV Line
Contingency: Albion Rd - Wixom 138 kV
Loading: 125.7% (Rating: 227 MVA)
DFAX: 29.4%
Pre-project loading: 73.9%
25. Facility: Ubc S Camps - Henry Crk 138 kV Line
Contingency: Albion Rd - Platte City Substation 138 kV
Loading: 124.7% (Rating: 182 MVA)
DFAX: 13.3%
Pre-project loading: 95.5%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Skeena - Pin Hook 138 kV	Rebuild line with higher-capacity conductor	\$8,728,000	56% (\$4,935,000)
Rapids Energy Ctr - Casino 138 kV	Rebuild line with higher-capacity conductor	\$3,474,000	74% (\$2,552,000)
Capbell Jct - Rapids Energy Ctr 138 kV	Rebuild line with higher-capacity conductor	\$2,651,000	74% (\$1,947,000)
Casno - Barnabie Cable Sta 138 kV	Rebuild line with higher-capacity conductor	\$6,790,000	74% (\$4,988,000)
Wixom - Albion Rd Substation 138 kV	Rebuild line with higher-capacity conductor	\$3,367,000	72% (\$2,417,000)
Skeena - Terry Co 138 kV line	Rebuild line with higher-capacity conductor	\$10,422,000	77% (\$8,018,000)
Cedar Tap - Terry Co 138 kV line	Rebuild line with higher-capacity conductor	\$9,778,000	77% (\$7,522,000)
Pin Hook - Anson North 138 kV L	Rebuild line with higher-capacity conductor	\$13,745,000	56% (\$7,771,000)
Barnabie Cable Sta - Moss Bluff 138 kV	Rebuild line with higher-capacity conductor	\$10,249,000	98% (\$10,025,000)
Albion Rd - Platte City 138 kV Line	Rebuild line with higher-capacity conductor	\$16,710,000	72% (\$11,978,000)
C-33 Doe - Platte City 138 kV	Rebuild line with higher-capacity conductor	\$4,284,000	72% (\$3,071,000)
Sand Springs - C-33 Doe 138 kV	Rebuild line with higher-capacity conductor	\$10,157,000	72% (\$7,281,000)
Longdon - Barnabie Cable Sta Substation 138 kV line	Rebuild line with higher-capacity conductor	\$7,087,000	100% (\$7,087,000)
Glendale S Tap 6 - Campbell Junction 138 kV	Rebuild line with higher-capacity conductor	\$3,743,000	83% (\$3,097,000)
Glendale S Tap 6 - Biscoe 138 kV	Rebuild line with higher-capacity conductor	\$3,081,000	83% (\$2,549,000)
Biscoe - Anson N 138 kV L	Rebuild line with higher-capacity conductor	\$5,194,000	83% (\$4,298,000)
Henry Crk - Central Campus Unm 138 kV	Rebuild line with higher-capacity conductor	\$12,707,000	78% (\$9,964,000)
Lovett - Idaho Falls Power 138 kV Line	Rebuild line with higher-capacity conductor	\$6,182,000	88% (\$5,462,000)
Wec Substation - Wellington 138 kV	Rebuild line with higher-capacity conductor	\$28,521,000	86% (\$24,440,000)
Lovett - Barnabie Cable Sta 138 kV Line	Rebuild line with higher-capacity conductor	\$24,780,000	95% (\$23,557,000)
Idaho Falls Power - Cedar Tap 138 kV	Rebuild line with higher-capacity conductor	\$6,229,000	88% (\$5,504,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
St Joseph - Chestnut Substation 138 kV line	Rebuild line with higher-capacity conductor	\$9,246,000	38% (\$3,532,000)
Central Campus Unm - Albion Road 138 kV Line	Rebuild line with higher-capacity conductor	\$4,884,000	78% (\$3,830,000)
Davisille - N Appleton 138 kV Line	Rebuild line with higher-capacity conductor	\$6,045,000	38% (\$2,309,000)
Ubc S Campus - Henry Crk 138 kV	Rebuild line with higher-capacity conductor	\$8,336,000	78% (\$6,536,000)
	Total allocated to project		\$174,670,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.